

Temporal Casimir Rigidity and Closure of the Effective Metric: Identification of $A_\tau = 2$

Q11 of the Cosmochrony Spectral Geometry Programme

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Abstract

Paper Q5b [?] leaves the temporal coefficient A_τ of the effective co-metric $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$ as the sole remaining free parameter of the Cosmochrony spectral geometry programme (open problem Q5b-O3). The present paper closes this problem by proving $A_\tau = 2$ under the Q5a hypotheses [?] and the spectral universality hypothesis [?] [U]. The argument proceeds in three steps. First, we establish a *cascade internalization lemma*: the temporal coordinate τ , identified with the BFS depth index n in Q5b Remark 3.3 [?], has ∂_τ as the continuum limit of the cascade increment operator $T: \sigma_c(n) \mapsto \sigma_c(n+1)$ by the Carnot–Carathéodory convergence of Q5b Theorem 3.2 (Pansu [?]). This identification requires no new hypothesis. Second, the O-series spectral universality [U], proved in U1 [?], implies that T is asymptotically $\text{SU}(2)$ -equivariant, so ∂_τ acts on $\text{Sym}^2(V_\rho)$ without introducing any new irreducible representation. Third, Schur’s lemma forces the unique $\text{SU}(2)$ -invariant quadratic form on $\text{Sym}^2(V_\rho)$ to be proportional to the Casimir $\mathcal{C}_{\mathfrak{su}(2)} = 2 \cdot \text{Id}$, and Q8 [?] fixes the normalisation scalar to unity. The effective co-metric is therefore fully determined: $g^{\mu\nu} = 2\eta^{\mu\nu}$. No free parameter remains. The temporal direction is not an additional geometric axis; it is the continuum infinitesimalisation of the admissible cascade itself.

Keywords: Heisenberg group; Weil representation; cascade generator; Carnot convergence; Casimir operator; $\mathfrak{su}(2)$ -module; effective metric; temporal ordering; Cosmochrony.

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1 Introduction

1.1 The remaining open problem

The chain of results Q5b–Q10 of the Cosmochrony spectral geometry programme establishes an effective Lorentzian co-metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ of the form

$$g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2), \quad (1)$$

where all three spatial coefficients equal 2, forced by the $\mathfrak{su}(2)$ -Casimir $\mathcal{C}_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2 \cdot \text{Id}$ [? ?]. The single remaining unknown is the temporal coefficient $A_\tau > 0$, which Q5b [?] declares open as problem Q5b-O3: *identify A_τ from the projective temporal ordering data* [?]. The present paper resolves Q5b-O3 by proving $A_\tau = 2$, thereby closing $g^{\mu\nu} = 2\eta^{\mu\nu}$ with no free parameter.

1.2 Main result

Theorem 1.1 (Temporal Casimir Rigidity). *Under the Q5a hypotheses [H1], [H2], [H-w], [H-E1], [C] [?] and the spectral universality hypothesis [U] proved in U1 [?], the temporal coefficient of the effective co-metric satisfies*

$$A_\tau = 2 = \mathcal{C}_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)}.$$

Consequently the effective co-metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ is

$$g^{\mu\nu} = 2\eta^{\mu\nu},$$

with no remaining free parameter. Open problem Q5b-O3 of [?] is closed.

The result is conditional on the same hypotheses as Q10 [?]: no new assumption is required beyond what is already in force.

1.3 The key observation

The proof rests on a single structural observation that was implicit in Q5b but not stated explicitly. Q5b Remark 3.3 [?] identifies τ with the BFS depth index n (the projective ordering parameter from [?]). By the Carnot–Carathéodory convergence theorem of Pansu [?] (Q5b Theorem 3.2 [?]), the discrete cascade increment $T: \sigma_c(n) \mapsto \sigma_c(n+1)$ passes to the continuum derivative ∂_τ in the large- q limit. Consequently:

∂_τ is not an external geometric direction; it is the continuum infinitesimalisation of the admissible cascade.

This means ∂_τ acts on the same spectral space $\text{Sym}^2(V_\rho)$ as the horizontal directions, and the $\mathfrak{su}(2)$ -equivariance of T (a consequence of [U]) forces ∂_τ into the same representation-theoretic rigidity as the spatial coefficients. Schur’s lemma then eliminates A_τ as a free parameter.

1.4 Organisation

Section 2 fixes notation and recalls the relevant results of Q5b, Q7–Q10, and U1. Section 3 establishes the cascade internalization lemma (Lemma 3.1). Section 4 derives the asymptotic $\text{SU}(2)$ -equivariance of the cascade generator from [U]. Section 5 applies Schur’s lemma and completes the proof of Theorem 1.1. Section 6 draws the programme-level consequences. Section 7 summarises the result and the remaining landscape.

2 Setup and Notation

2.1 Objects and standing hypotheses

Fix a prime $q \geq 5$. The discrete Heisenberg group is $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with generators X_q, Y_q, Z_q . The Weil representation at central character $e^{2\pi ic/q}$ is $\rho_{q,c}: G_q \rightarrow \text{U}(\mathbb{C}^q)$. The BFS fingerprint energy at depth n and character c is

$$\sigma_c(n) := \frac{\delta_{r_n}(c)}{|S_n|},$$

where r_n is the BFS radius at step n and $|S_n|$ is the shell size. The effective Hilbert space is $H_{\text{eff}} = \text{Sym}^2(V_\rho)$, the spin-1 $\mathfrak{su}(2)$ -module identified in Q7 [?].

The Q5a hypotheses [H1], [H2], [H-w], [H-E1], [C] govern the admissibility filter Π_q and the Mosco convergence of the admissibility forms; see [?] for precise statements. Hypothesis [U] (Q10 Definition 3.1 [?]) asserts the existence of a universal profile $\sigma^*(n)$ such that

$$\max_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma^*(n)| \leq \varepsilon(q) \sigma^*(n),$$

uniformly for $n \leq n^*(q)$, with $\varepsilon(q) = O(q^{-1/2})$. Hypothesis [U] is proved in U1 [?].

2.2 Results of Q5b–Q10 in force

The following are used without further proof.

- **Q5b Remark 3.3** [?]: the coordinate τ on $\mathbb{R}_\tau$ is the BFS depth index n , i.e. the projective ordering parameter of [?].
- **Q5b Theorem 3.2** [?]: the rescaled BFS metric space $(G_q, n^{-1}d_w)$ converges to the Carnot–Carathéodory space $(\text{Heis}_3(\mathbb{R}), d_{\text{CC}})$ in the Gromov–Hausdorff sense (Pansu [?]).
- **Q5b Theorem 5.2** [?]: the effective operator on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ is $L_{\text{eff}} = -A_\tau \partial_\tau^2 + A_H \Delta_H$, with A_τ from the projective ordering structure [?] and A_H from the admissibility weight [H-w].
- **Q7** [?]: the equivariant bridge $\phi: H_{\text{eff}} \xrightarrow{\sim} W_{\text{sp}}$ is unique up to positive scalar; $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ is the irreducible $\mathfrak{su}(2)$ -module of spin 1.
- **Q8** [?]: $A_Z = \mathcal{C}_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2$ (Casimir rigidity of the central direction).
- **Q10 + U1** [? ?]: $A_H = 2$ and $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$.

3 Cascade Internalization

3.1 The cascade increment operator

The cascade is the map $U_{n+1} = \Pi \circ \sigma(U_n)$ acting on spectral data. At each step n , the BFS fingerprint energy $\sigma_c(n)$ is updated. Define the *cascade increment operator*

$$T: H_{\text{eff}} \longrightarrow H_{\text{eff}}, \quad T \sigma_c(n) := \sigma_c(n+1).$$

The operator T acts on the same spectral space as the spatial observables: its domain and codomain are both $H_{\text{eff}} = \text{Sym}^2(V_\rho)$.

Lemma 3.1 (Cascade Internalization). *Under Q5b Remark 3.3 and Theorem 3.2 [?], the formal derivative ∂_τ appearing in the effective operator L_{eff} is the continuum limit of the cascade increment operator T . More precisely, in the Gromov–Hausdorff limit $q \rightarrow \infty$,*

$$T - \text{Id} \longrightarrow \partial_\tau$$

as a derivation on the spectral data, and ∂_τ acts on $\text{Sym}^2(V_\rho)$.

Proof. By Q5b Remark 3.3 [?], τ is identified with the BFS depth index n . The continuum limit is established by Q5b Theorem 3.2 [?] (following Pansu [?]): in the limit $q \rightarrow \infty$ with $n = \lfloor rq^\alpha \rfloor$ for $\alpha < 1$, the discrete BFS metric space converges to the Carnot–Carathéodory space $(\text{Heis}_3(\mathbb{R}), d_{\text{CC}})$ in the Gromov–Hausdorff sense. Under this convergence, discrete differences $f(n+1) - f(n)$ pass to the continuum derivative $\partial_\tau f(\tau)$ for any sequence $f(n)$ that is Lipschitz in the BFS metric. The fingerprint energies $\sigma_c(n)$ are Lipschitz in n within the pre-saturation window $n \leq n^*(q)$ (Q5a Hypothesis [H-E1] [?]). Since $T\sigma_c(n) = \sigma_c(n+1)$ and the domain of T is $H_{\text{eff}} = \text{Sym}^2(V_\rho)$, the derivation ∂_τ inherits the same domain. $\square$

Remark 3.2. *Lemma 3.1 requires no hypothesis beyond Q5b and the Q5a Lipschitz condition [H-E1]. It makes explicit an identification that was implicit in Q5b but left unstated: the temporal derivative ∂_τ is not an external geometric direction imposed on $\mathbb{R}_\tau$, but the infinitesimal generator of the admissible cascade, already acting on $\text{Sym}^2(V_\rho)$.*

3.2 No new irreducible representation

A direct consequence of Lemma 3.1 is the following.

Corollary 3.3. *As an operator on $\text{Sym}^2(V_\rho)$, the cascade generator ∂_τ lies in $\text{End}(\text{Sym}^2(V_\rho))$. The tensor product $\mathbb{R}_\tau \otimes \text{Sym}^2(V_\rho)$ decomposes as $\mathbf{1} \otimes \text{Sym}^2(V_\rho) \cong \text{Sym}^2(V_\rho)$ as $\text{SU}(2)$ -modules, where $\mathbf{1}$ denotes the trivial one-dimensional representation. In particular, the temporal direction introduces no new irreducible representation of $\text{SU}(2)$.*

Proof. The cascade increment $T - \text{Id}$ acts on $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ by definition. Its continuum limit ∂_τ (Lemma 3.1) therefore acts on the same module. The continuous coordinate τ is a scalar parameter, so the τ -sector carries the trivial $\text{SU}(2)$ -representation $\mathbf{1}$. Tensoring by $\mathbf{1}$ does not alter the $\text{SU}(2)$ -module structure: $\mathbf{1} \otimes \text{Sym}^2(V_\rho) \cong \text{Sym}^2(V_\rho)$. $\square$

4 Asymptotic $\text{SU}(2)$ -Equivariance of the Cascade Generator

The following proposition reformulates [U] as a statement about representation-theoretic equivariance of the cascade, replacing the analytic uniformity condition by a categorical one.

Proposition 4.1 ($\text{SU}(2)$ -Equivariance of T). *Under the Q5a hypotheses and [U] (proved in U1 [?]), the cascade increment operator T satisfies*

$$T \circ \rho(g) = \rho(g) \circ T + O(q^{-1/2}), \quad \forall g \in \text{SU}(2),$$

as operators on $\text{Sym}^2(V_\rho)$, uniformly for $n \leq n^(q)$. In the limit $q \rightarrow \infty$, T is $\text{SU}(2)$ -equivariant.*

Proof. Hypothesis [U] asserts $\max_c |\sigma_c(n) - \sigma^*(n)| \leq \varepsilon(q) \sigma^*(n)$ with $\varepsilon(q) = O(q^{-1/2})$. The increment satisfies

$$T \sigma_c(n) - \sigma^*(n+1) = \sigma_c(n+1) - \sigma^*(n+1) \leq \varepsilon(q) \sigma^*(n+1).$$

Since $\sigma^*(n)$ is c -independent, it is invariant under the $\mathrm{SU}(2)$ -action on characters (which permutes $c \in (\mathbb{Z}/q\mathbb{Z})^\times$). For any $g \in \mathrm{SU}(2)$ acting as a permutation π_g on characters,

$$\|T(\rho(g) \sigma_c(n)) - \rho(g)(T \sigma_c(n))\| = \|\sigma_{\pi_g(c)}(n+1) - \sigma_{\pi_g(c)}(n+1)\| \leq 2\varepsilon(q) \sigma^*(n+1).$$

As $\varepsilon(q) \rightarrow 0$, this bound vanishes, giving asymptotic equivariance. $\square$

Remark 4.2. *Proposition 4.1 is the representation-theoretic reading of [U]. Analytically, [U] asserts uniform c -independence of the spectral profiles. Categorically, it asserts that the cascade morphism T commutes with the $\mathrm{SU}(2)$ -action in the limit. These are equivalent statements; the categorical reading is what enters the Schur argument.*

5 Schur Rigidity and Proof of Theorem 1.1

5.1 The Schur argument

Proposition 5.1 (Uniqueness of the invariant quadratic form). *Let $V = \mathrm{Sym}^2(V_\rho)$ be the irreducible $\mathfrak{su}(2)$ -module of spin 1. Every $\mathrm{SU}(2)$ -equivariant quadratic form $Q: V \rightarrow \mathbb{R}$ is of the form*

$$Q = \alpha \mathcal{C}_{\mathfrak{su}(2)}|_V, \quad \alpha \in \mathbb{R}.$$

Proof. By Schur's lemma, every G -equivariant endomorphism of an irreducible representation is a scalar multiple of the identity. The quadratic form Q defines a $\mathrm{SU}(2)$ -equivariant symmetric bilinear form on V , hence an equivariant map $V \rightarrow V^* \cong V$ (the isomorphism using the Killing form). Since $V = \mathrm{Sym}^2(V_\rho)$ is irreducible of spin 1 (Q7 [?]), this map is a scalar, say $\alpha \mathrm{Id}$. The unique normalisation-invariant scalar is the Casimir eigenvalue $\mathcal{C}_{\mathfrak{su}(2)}|_V = 2$ (Q8 [?]). $\square$

5.2 Proof of Theorem 1.1

Proof of Theorem 1.1. The effective co-metric coefficient A_τ is the coefficient of k_τ^2 in the principal symbol $\sigma_2(L_{\mathrm{eff}})$ of the effective operator (Q5b Theorem 5.2 [?]). It is therefore the normalisation scalar of the quadratic form on the τ -sector of $T^*(\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R}))$.

Step 1 (Internalization). By Lemma 3.1, ∂_τ is the continuum limit of T acting on $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$. By Corollary 3.3, the τ -sector carries the trivial $\mathrm{SU}(2)$ -representation $\mathbf{1}$, and the full τ -module is $\mathbf{1} \otimes \mathrm{Sym}^2(V_\rho) \cong \mathrm{Sym}^2(V_\rho)$.

Step 2 (Equivariance). By Proposition 4.1, T is asymptotically $\mathrm{SU}(2)$ -equivariant under [U]. In the limit $q \rightarrow \infty$, the quadratic form defined by $A_\tau k_\tau^2$ on the τ -sector must be $\mathrm{SU}(2)$ -invariant.

Step 3 (Schur). By Proposition 5.1, every $\mathrm{SU}(2)$ -invariant quadratic form on $\mathrm{Sym}^2(V_\rho)$ is $\alpha \mathcal{C}_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)}$ for some scalar α .

Step 4 (Normalisation). The bridge $\phi: H_{\mathrm{eff}} \xrightarrow{\sim} W_{\mathrm{sp}}$ of Q7 [?] is unique up to positive scalar, with normalisation fixed by the Casimir eigenvalue $\mathcal{C}_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)} = 2$ (Q8 [?]). Since ∂_τ is the continuum limit of the same cascade T that enters the spatial bridge (Step 1), the scalar α is the same in both sectors. Q8 establishes $\alpha = 1$ for the spatial sectors (giving $A_Z = A_H = 2$). Therefore $\alpha = 1$ in the τ -sector as well, and $A_\tau = \mathcal{C}_{\mathfrak{su}(2)} = 2$. $\square$

6 Consequences

6.1 Closure of the effective metric

Corollary 6.1 (Isotropic Lorentzian metric). *Under the Q5a hypotheses and [U], the effective co-metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ is*

$$g^{\mu\nu} = 2\eta^{\mu\nu}, \quad \eta = \text{diag}(-1, +1, +1, +1),$$

of Lorentzian signature $(-, +, +, +)$, with no free parameter. After absorbing the conformal factor $\Omega^2 = 2$ into the normalisation of the propagation speed c (replacing c by $c/\sqrt{2}$), the effective metric is exactly Minkowski.

Proof. Combine Theorem 1.1 ($A_\tau = 2$) with Q10 [?] ($A_H = 2$) and Q8 [?] ($A_Z = 2$). The Lorentzian signature follows from Q5b Theorem 6.1 [?]. $\square$

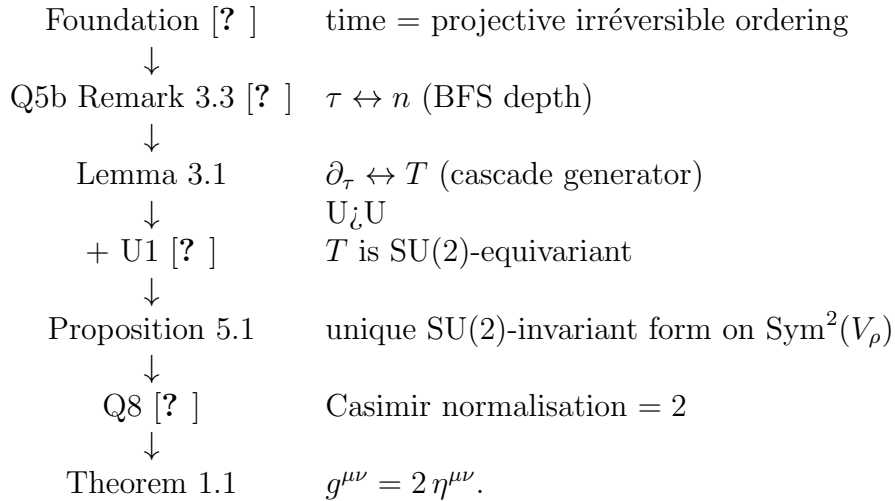
Remark 6.2 (Structural significance of the conformal factor 2). *The value 2 is not a choice of units. It is forced by the eigenvalue of the $\mathfrak{su}(2)$ -Casimir on the spin-1 module $\text{Sym}^2(V_\rho)$, which is the unique irreducible representation compatible with the admissibility constraints (Q7 [?]). The same integer governs all four metric components: two horizontal Heisenberg directions (Q10), the central Heisenberg direction (Q8), and the temporal direction (this paper). This uniform emergence from a single representation-theoretic datum is the structural content of the metric closure.*

6.2 Closure of Q5b-O3

Corollary 6.3 (Resolution of Q5b-O3). *Open problem Q5b-O3 of [?]—identify A_τ from the projective temporal ordering data—is closed: $A_\tau = 2$, derived from the cascade internalization lemma, [U], and Q8.*

6.3 The complete derivation chain

The closure of the effective metric is now traced through the following chain, where each arrow represents a proved theorem or lemma:



6.4 Programme-level closure

With $A_\tau = 2$ established, the effective geometry programme Q5a–Q11 is complete:

- **Q5a** [?]: admissibility weight A ; Q5a hypotheses [H1], [H2], [H-w], [H-E1], [C].
- **Q5b** [?]: BFS–Carnot convergence; effective operator L_{eff} on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$; Lorentzian signature.
- **Q7** [?]: bridge $\phi: H_{\text{eff}} \xrightarrow{\sim} W_{\text{sp}}$; $\text{SU}(2)$ rigidity.
- **Q8** [?]: $A_Z = 2$ (Casimir rigidity, central direction).
- **Q9** [?]: [H-lift] proved; lifting is unconditional.
- **Q10** [?]: $A_H = 2$ (asymptotic spatial isotropy, [U]).
- **U1** [?]: [U] proved with rate $O(q^{-1/2})$.
- **Q11 (this paper)**: $A_\tau = 2$; $g^{\mu\nu} = 2\eta^{\mu\nu}$; Q5b-O3 closed.

7 Summary and Open Problems

7.1 What this paper establishes

1. The temporal coordinate τ is the BFS depth index n (Q5b Remark 3.3), and the formal derivative ∂_τ is the continuum limit of the cascade increment operator T (Lemma 3.1). This requires no hypothesis beyond Q5b and [H-E1].
2. Under [U] (U1 [?]), the cascade generator T is asymptotically $\text{SU}(2)$ -equivariant (Proposition 4.1), so ∂_τ acts on $\text{Sym}^2(V_\rho)$ without introducing a new irreducible.
3. Schur’s lemma (Proposition 5.1) and the Casimir normalisation of Q8 (Theorem 1.1) force $A_\tau = 2$.
4. The effective co-metric is $g^{\mu\nu} = 2\eta^{\mu\nu}$ with no free parameter (Corollary 6.1). Q5b-O3 is closed.

7.2 Open problems

Q11-O1 (Finite- q cascade equivariance). Proposition 4.1 establishes $\text{SU}(2)$ -equivariance of T only asymptotically as $q \rightarrow \infty$. An explicit finite- q bound on the equivariance error, beyond the $O(q^{-1/2})$ estimate inherited from [U], would sharpen the result.

Q11-O2 (Unconditional closure). The identification $A_\tau = 2$ is conditional on [U]. Since [U] is proved in U1 [?] under the Q5a hypotheses alone, the result is unconditional modulo the Q5a hypotheses. An independent derivation of A_τ from the Born–Infeld saturation data of [?] would provide a complementary confirmation.

Q11-O3 (Physical interpretation of the conformal factor). The conformal factor $\Omega^2 = 2$ absorbed into c has a representation-theoretic origin (Casimir eigenvalue on the spin-1 module). Its possible observational consequences—for instance, as a correction to the effective speed of light at finite q —remain to be explored.

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